Sentiment Analysis using Transformer Models – Hugging Face

step 1: Environment Setup

```
!pip install transformers
!pip install datasets
!pip install torch
!pip install scikit-learn
```

```
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transfo
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfac
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0-
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hu
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.24.3)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets) (4.67.3)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (1.5.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datas
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datas
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfac
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0-
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->d
```

step 2: Data Loading

```
from datasets import load_dataset

dataset = load_dataset("imdb")

train_data = dataset['train']
test_data = dataset['test']
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
README.md:        7.81k/?  [00:00<00:00,  425kB/s]

plain_text/train-00000-of-00001.parquet:  100%                                    21.0M/21.0M  [00:00<00:00,  22.9MB/s]

plain_text/test-00000-of-00001.parquet:  100%                                    20.5M/20.5M  [00:00<00:00,  32.1MB/s]

plain_text/unsupervised-00000-of-00001.p(…):  100%                                    42.0M/42.0M  [00:00<00:00,  164MB/s]

Generating  train  split:  100%                                    25000/25000  [00:00<00:00,  90293.64  examples/s]

Generating  test  split:  100%                                    25000/25000  [00:00<00:00,  93295.39  examples/s]

Generating  unsupervised  split:  100%                                    50000/50000  [00:00<00:00,  127144.68  examples/s]
```

Step 3: Text Preprocessing

Clean text and prepare tokenizer.

```
from transformers import DistilBertTokenizer

tokenizer = DistilBertTokenizer.from_pretrained('distilbert-base-uncased')
```

```
tokenizer_config.json:  100%                                    48.0/48.0  [00:00<00:00,  1.47kB/s]

vocab.txt:        232k/?  [00:00<00:00,  3.84MB/s]

tokenizer.json:        466k/?  [00:00<00:00,  5.70MB/s]
```

Tokenization:

```
def tokenize(example):
    return tokenizer(example['text'], padding='max_length', truncation=True)

train_data = train_data.map(tokenize, batched=True)
test_data = test_data.map(tokenize, batched=True)
```

```
Map:  100%                                    25000/25000  [00:28<00:00,  926.76  examples/s]

Map:  100%                                    25000/25000  [00:33<00:00,  659.96  examples/s]
```

Step 4: Vocabulary Creation & Encoding

The tokenizer converts text into tokens and assigns each token a numerical ID.

```
r_str = "I love this movie"

# Tokens
# ["I", "love", "this", "movie"]

# IDs
# [101, 1045, 2293, 2023, 3185, 102]
```

Step 5: Train-Test Split

The dataset already provides:

25,000 training samples

25,000 testing samples

Step 6: Create PyTorch Dataset

```
import torch

train_data.set_format(type='torch', columns=['input_ids','attention_mask','label'])
test_data.set_format(type='torch', columns=['input_ids','attention_mask','label'])
```

Step 7: Build transformer model

```
from transformers import DistilBertForSequenceClassification

model = DistilBertForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)
```

```
config.json: 100%                                                    483/483 [00:00<00:00, 22.0kB/s]

model.safetensors: 100%                                              268M/268M [00:03<00:00, 97.5MB/s]

Loading weights: 100%                          100/100 [00:00<00:00, 362.14it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_projector.bias     | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_layer_norm.weight  | UNEXPECTED |
classifier.bias          | MISSING    |
pre_classifier.bias      | MISSING    |
pre_classifier.weight    | MISSING    |
classifier.weight        | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream t
```

Step 8: Model Training

```
import numpy as np
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
from transformers import Trainer, TrainingArguments

# Define compute_metrics function
def compute_metrics(pred):
    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)
    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds, average='binary')
    acc = accuracy_score(labels, preds)
    return {
        'accuracy': acc,
        'f1': f1,
        'precision': precision,
        'recall': recall
    }

# Define training arguments
training_args = TrainingArguments(
    output_dir="./results",
    num_train_epochs=3, # Reduced epochs for quicker demo
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    warmup_steps=500,
    weight_decay=0.01,
    logging_dir="./logs",
    logging_steps=10
    # evaluation_strategy="epoch" # Removed evaluation_strategy as it caused an error
)

# Initialize Trainer
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_data, # Corrected variable name
    eval_dataset=test_data, # Corrected variable name
    # tokenizer=tokenizer, # Removed tokenizer as it's not a valid argument for Trainer
    compute_metrics=compute_metrics
)

# Start training
trainer.train()
```

```
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD_LOGGING_DIR` instead.
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_371/1205956206.py in <cell line: 0>()
     31 # Initialize Trainer
     32 trainer = Trainer(
---> 33     model=model,
     34     args=training_args,
     35     train_dataset=train_data, # Corrected variable name

NameError: name 'model' is not defined
```

Step 9: Model Evaluation

```
trainer.evaluate()
```